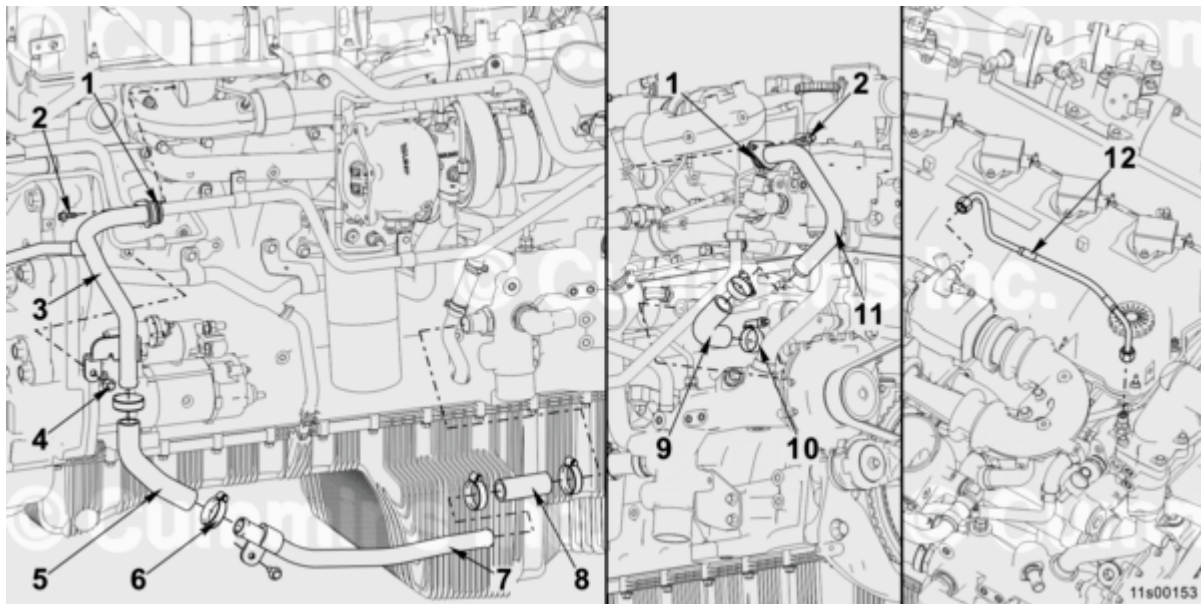


Trainings ▾

Exploded View



Exhaust Gas Recirculation (EGR) Cooler Coolant Lines Exploded View

1. O-ring - quantity 2
2. Capscrew - quantity 2
3. EGR cooler water return tube - quantity 1
4. Capscrew - quantity 1
5. Hose - quantity 1
6. Clamp - quantity 4
7. EGR cooler water return tube - quantity 1
8. Hose - quantity 1
9. Hose - quantity 1
10. Clamp - quantity 2
11. EGR cooler water supply tube - quantity 1
12. EGR cooler coolant vent line - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- Air Handling Clean Care Kit, Part Number 4919799

Additional Service Items

- No additional service items required.

General Information

EGR coolant lines can be either individual or connected in series with retarder plumbings.

Use the following procedure for more information on the series EGR coolant lines. Refer to Procedure 008-152 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html>)

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

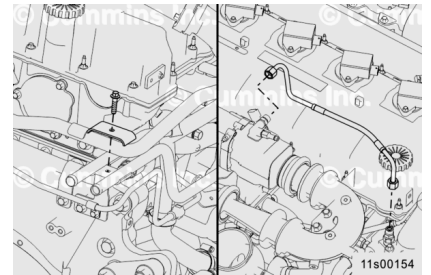
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html>)
- Remove the retarder plumbing. Refer to Procedure 008-152 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html>)
- Remove any Original Equipment Manufacturer (OEM) air cleaner and charge air cooler plumbing needed to remove the coolant lines. See equipment manufacturer service

information. Use protective caps from the Air Handling Clean Care Kit, Part Number 4919799, to cover the connection points.

Remove

⚠ CAUTION ⚠

Be sure all the components are clean and free of debris that can cause damage to o-ring seals.



Remove the capscrew securing the evaporator tube.

Disconnect the EGR cooler coolant vent line from the coolant thermostat housing and the EGR cooler.

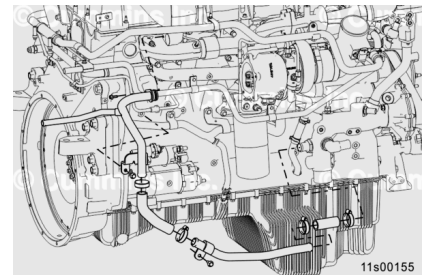
Remove the clamps and hoses on the EGR cooler coolant return tube.

Remove the EGR cooler coolant return tube mounting capscrews at the EGR cooler coolant outlet.

Remove capscrews from cylinder block and retarder plumbing bracket.

Remove the EGR cooler coolant return tube from the engine.

Remove and discard the o-ring.

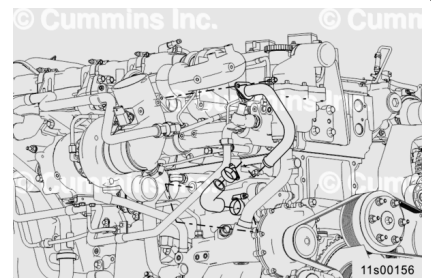


Remove the clamps and hoses on the EGR cooler coolant supply tube.

Remove the EGR cooler coolant supply tube mounting capscrew at the EGR coolant inlet.

Remove the EGR cooler coolant supply tube from the engine.

Remove and discard the o-ring.



Clean and Inspect for Reuse

⚠ WARNING ⚠

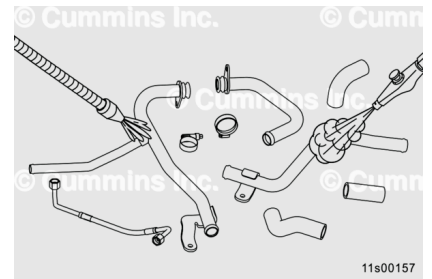
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the EGR cooler coolant vent line and the EGR cooler coolant supply and return tubes with safety solvent.

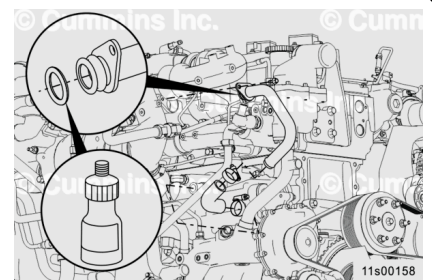
Dry with compressed air.



Install

⚠ CAUTION ⚠

Do not use petroleum products on the EGR cooler coolant line o-rings. The o-rings will swell and cause the connection to leak.



Install a new o-ring on the EGR cooler coolant supply tube.

Lubricate the o-ring with clean engine coolant, soapy water, or vegetable oil to aid in installation.

Tighten the clamp to connect the EGR cooler coolant supply hose and EGR cooler coolant supply tube.

Tighten the clamp to connect the EGR cooler coolant supply hose and the connector on the oil cooler.

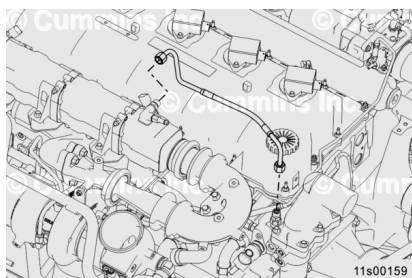
Attach the EGR cooler coolant supply tube into the EGR cooler coolant inlet.

Tighten the mounting capscrew to the EGR cooler.

Torque Value: 8 n•m [71 in-lb]

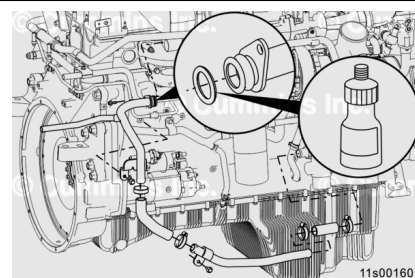
Install the EGR cooler coolant vent line to the coolant thermostat housing and the EGR cooler.

Torque Value: 27 n•m [239 in-lb]



⚠ CAUTION ⚠

Do not use petroleum products on the EGR cooler coolant line o-rings. The o-rings will swell and cause the connection to leak.



Install the o-ring on the EGR cooler coolant return tube.

Lubricate the o-ring with clean engine coolant, soapy water, or vegetable oil to aid in installation.

Attach the upper EGR cooler coolant return tube into the EGR cooler coolant outlet.

Tighten the mounting capscrew to the EGR cooler.

Torque Value: 24 n•m [212 in-lb]

Tighten the mounting capscrew securing the upper EGR cooler coolant return tube to cylinder block.

Torque Value: 47 n•m [35 ft-lb]

Tighten the mounting capscrew securing the lower EGR cooler coolant return tube to retarder plumbing bracket.

Torque Value: 47 n•m [35 ft-lb]

Tighten the clamps to connect the two EGR cooler coolant return hoses.

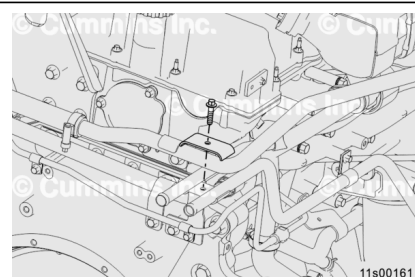
Torque Value: 5 n•m [44 in-lb]

Tighten the clamps to connect the lower EGR cooler coolant return hose and water inlet connection.

Torque Value: 5 n•m [44 in-lb]

Install the capscrew securing the evaporator tube.

Torque Value: 24 n•m [212 in-lb]



Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Remove the protective caps from the intake piping.
- Install the retarder plumbing. Refer to Procedure 008-152 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html)
- Install the intake piping. See equipment manufacturer service information.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.